

Mixed Reality Environment for Testing Automated Vehicle and Pedestrian Interaction

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Figure 1: Automatic Emergency Braking testing in the Mixed Reality Environment, including the Virtual Environment (top), the Pedestrian with trackers and head mounted display (bottom left) and the Dynamic Vehicle-in-the-Loop (bottom right)

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ABSTRACT

The test and development of Automated Driving Systems is usually realized by scenario based testing or virtual testing environments.

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These methods apply artificial targets to trigger the safety critical functions under specific predefined scenarios, as the NCAP or IIHS test catalogues. Despite having a good reproducibility, these approaches hardly permit the evaluation of new interaction concepts like external Human-Machine Interfaces (eHMI), since the interaction between real users and the vehicle cannot be realistically reproduced without risks to the participants. The novel Mixed Reality Test Environment (MiRE) overcomes this limitation by the integration of Virtual Reality (VR) technologies, Dynamic Vehicle-in-the-Loop (DynViL) and the Virtual Environment. In MiRE the movement and positioning of the vehicles and the VRUs are tracked in real time and reproduced in the virtual environment. Synthetic data from the virtual environment is generated to stimulate the vehicle and the human participant, enabling a safe interaction between both entities.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality**.

KEYWORDS

Automated Driving Systems; Test Procedures; Vehicle-in-the-Loop; External Human-Machine Interfaces; Sensor stimulation

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1 INTRODUCTION

The development of Automated Driving Systems requires an extensive test chain to guarantee the safe operation of the system [1]. This development is supported by virtual testing and the final validation is mainly conducted on test tracks under test scenarios predefined in NCAP [5] or IIHS [7] test catalogues. By the increase of the vehicle automation level and the consequently higher decision capacities of the vehicle, scenarios that are more complex need to be evaluated. It includes the interaction between the Vulnerable Road Users (VRUs) and the automated system, since unexpected movements of the VRU can provoke misbehaviors of the automated system. A pedestrian might for example get startled by the approaching vehicle, which might lead to unpredictable movement of the pedestrian. This can, in return influence the accuracy of the automated system's perception of a situation. The current test approach, however, does not permit to evaluate the interaction of the system with VRUs without inserting high risk to the participants. To overcome this limitation, the Mixed Reality Test Environment (MiRE) integrates an overarching architecture, which allows the integration of a Dynamic Vehicle-in-the-Loop, a VRU in a VR-environment and the simulated environment into one experimental environment. It enables the virtual interaction between the participant in real time, while they move in separated and safe spaces.

2 RELATED WORK

The development of automated driving vehicles has been accelerated by the integration of simulators and the driving system, using technologies as Hardware-in-the-Loop (HiL) or Vehicle-in-the-Loop (ViL). The testing benches include concepts from unique real components as sensors [8] until the whole driving system [9], while the missing components and environment are simulated. These methods use mathematical models to describe the dynamic behaviour of the traffic participants, which have limitations due to the physical simplifications. Aiming to reproduce the real behaviour of the vehicle in the virtual environment, novel ViL approaches integrate the real test vehicle into a virtual test scenario. In this case, the vehicle drives on an empty test track, while the environment surroundings are feed from the simulation into the vehicle system [3, 11, 12]. In the proposed architectures, simplified sensor models provide the synthetic data injected into the vehicle system. Due to that, the perception algorithms are not considered and the interaction between external road users and the vehicle can be evaluated just in limited cases.

An important aspect of automated driving is the interaction between VRUs and automated vehicles. The behavior of automated vehicles in situations where they interact with other road users can significantly impact their acceptance. Research in this area has especially focused on external human-machine interfaces for AVs. However, the evaluation of these interfaces is currently conducted in simulators or as Wizard-of-Oz experiments [4, 6]. These experiments can't consider the interaction of VRU's behavior and the underlying automated driving functions of the vehicle. Another area of research is investigating AR overlays for drivers and passengers of AVs, to inform them about the automation's current understanding of the situation and prevent "black-box" decisions [10, 13, 17]. In experiments, these augmentations are currently based on optimized information of the vehicle's surroundings and thus mostly show an idealistic representation of this technology's capabilities. Similarly, cooperative approaches to automated driving have been investigated in simulator environments [14–16]. These studies looked at situations, in which the driver of an automated vehicle can work together in ambiguous situations, to increase the efficiency of the drive. Due to the nature of these simulator experiments, the interaction between other road users and the AV, assisted by the driver in a cooperative manner can only be simulated with a limited level of realism. A simplified demonstration of the Mixed Reality Environment including a pedestrian and a pure simulated automated vehicle is to be presented at the conference Mensch und Computer 2021.

3 MIXED REALITY TESTING APPROACH

The MiRE consist of a modular architecture, which permits the integration of diverse entities and simulators simultaneously. The system is based on standardized interfaces provided by the Open Simulation Interface (OSI) and a global knowledge base, where the state of all participants is shared. This approach enables the participants to operate in different areas and to exchange the positioning data. As shown in Fig. 1 in this demonstration two external entities

interact on the virtual environment, a Pedestrian with Virtual Reality equipments and a vehicle connect with the simulator using the Dynamic Vehicle-in-the-Loop (DynViL) approach.

The DynViL is composed by a complete automated driving stack implemented on the Robot Operating System (ROS), including the perception, decision-making and the controlling units. These algorithms run on a prototyping ADAS platform separated from the vehicle under test, enabling a physical connection between simulator and ADAS platform and the consequent real time stimulation of sensors of the vehicle by the simulator. The control commands are remotely sent to the real vehicle on the test track, and driving robots control the vehicle. The state of the vehicle is obtained by an Inertial measurement unit (IMU) and GPS and is sent back to the ADAS platform to feed the controlling and decision-making algorithms. Besides that, a bridge between the ROS framework and the OSI communication permits the vehicle position exchange with the global knowledge base. In the virtual environment, a digital twin of the vehicle is positioned according to the received data. On the virtual vehicle the virtual sensors are integrated, which generated the raw data necessary to stimulate the vehicle's sensors. A direct data injection approach is used to feed the simulator image into the ADAS Platform. It consists of a Video Interface Box (ViB), which converts the RGB video to raw images compatible with the ADAS Platform and its specific serialized format. The data is finally recognized as a camera data by the ECU.

To allow for a realistic interaction between the VRU (pedestrian) and the vehicle a body-tracking system based on the HTC Vive HMD, combined with the Vive Trackers, similarly to [2] is implemented to track the pedestrian movements. For this system, seven trackers in combination with the HMD and Vive Controllers are used. Based on the positions of the trackers, which are placed on the hip, elbows, knees, and feet, the body pose can be determined using inverse kinematics. A virtual avatar in Unity 3D is then manipulated accordingly, to represent the pose of the real person. This process allows for an accurate representation of the pose and movement of a person in the simulation environment. The virtual environment is rendered and displayed on the HMD to stimulate the pedestrian. This allows a participant inside the VR simulation environment to see the approaching vehicle. The vehicle on the other hand is also able to perceive the pedestrian in the scenario and react to him.

The proposed approach provides the possibility to combine simple individual scenarios into a complex one. The overall scenario can be used to investigate the interaction of individual participants, such as automated vehicles and pedestrians. The physical separation allows investigating corner cases or misbehavior of VRUs without physically harming them. In addition to these possibilities, experiments can be performed by several automated vehicles interacting at a common environment. The same approach can be implemented to human drivers, which have to interact cooperatively with other drivers and automated vehicles. Therefore, multiple driving simulators can be extended with heads-up display or augmented reality systems, to evaluate the individual behavior of single drivers but also the interaction of multiple participants in a complex scenario. Since only parts of the entire scenario have to be displayed for the individual participants, scalability to complex scenarios with numerous participants is conceivable due to the distributed system design.

4 MIXED REALITY ENVIRONMENT DEMO-VIDEO

The video is a demonstration of the MiRE. The use case reproduces a test scenario based on the EURO NCAP catalogue for testing Automatic Emergency Braking (AEB) functions for VRU protection. In the demonstration, a camera based perception unit and an automated driving function are used to guide the DynViL straight at a specific velocity and to brake under safety critical scenarios. The pedestrian target usually used on Euro NCAP tests is substituted by a VR-pedestrian, which moves in a separated room. The pedestrian sees the virtual environment by the HMD and his movements are tracked and sent to the virtual environment in real time. In the video is possible to identify the reproduction of the movements of both entities inside the simulated environment and to observe the image detection in action. Furthermore, the limitation of pure simulated or real testing are explained and further development of the MiRE are presented.

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